

Ex. 11. Simplify: (i) $\frac{789 \times 789 \times 789 + 211 \times 211 \times 211}{789 \times 789 - 789 \times 211 + 211 \times 211}$ (ii) $\frac{658 \times 658 \times 658 - 328 \times 328 \times 328}{658 \times 658 + 658 \times 328 + 328 \times 328}$

Sol. (i) Given exp. = $\frac{(789)^3 + (211)^3}{(789)^2 - (789 \times 211) + (211)^2} = \frac{a^3 + b^3}{a^2 - ab + b^2}$, (where $a = 789$ and $b = 211$)
 $= (a + b) = (789 + 211) = 1000.$

(ii) Given exp. = $\frac{(658)^3 - (328)^3}{(658)^2 + (658 \times 328) + (328)^2} = \frac{a^3 - b^3}{a^2 + ab + b^2}$, (where $a = 658$ and $b = 328$)
 $= (a - b) = (658 - 328) = 330.$

Ex. 12. Simplify: $\frac{(893 + 786)^2 - (893 - 786)^2}{(893 \times 786)}.$

Sol. Given exp. = $\frac{(a + b)^2 - (a - b)^2}{ab}$ (where $a = 893, b = 786$) = $\frac{4ab}{ab} = 4.$

Ex. 13. Which of the following are prime numbers?

(i) 241

(ii) 337

(iii) 391

(iv) 571

Sol. (i) Clearly, $16 > \sqrt{241}.$

Prime numbers less than 16 are 2, 3, 5, 7, 11, 13.

241 is not divisible by any of them.

$\therefore$ 241 is a prime number.

(ii) Clearly, $19 > \sqrt{337}.$ Prime numbers less than 19 are 2, 3, 5, 7, 11, 13, 17.

337 is not divisible by any one of them.

$\therefore$ 337 is a prime number.

(iii) Clearly, $20 > \sqrt{391}.$ Prime numbers less than 20 are 2, 3, 5, 7, 11, 13, 17, 19.

We find that 391 is divisible by 17.

$\therefore$ 391 is not a prime number.

(iv) Clearly, $24 > \sqrt{571}.$ Prime numbers less than 24 are 2, 3, 5, 7, 11, 13, 17, 19, 23.

571 is not divisible by any one of them.

$\therefore$ 571 is a prime number.

Ex. 14. If Δ stands for the operation 'adding first number to twice the second number', then find the value of $(1 \Delta 2) \Delta 3.$

Sol. $(1 \Delta 2) \Delta 3 = (1 + 2 \times 2) \Delta 3 = 5 \Delta 3 = 5 + 2 \times 3 = 5 + 6 = 11.$

Ex. 15. Given that $1^2 + 2^2 + 3^2 + \dots + 10^2 = 385$, then find the value of $2^2 + 4^2 + 6^2 + \dots + 20^2.$

Sol. $2^2 + 4^2 + 6^2 + \dots + 20^2 = 2^2 (1^2 + 2^2 + 3^2 + \dots + 10^2) = 2^2 \times 385 = 4 \times 385 = 1540.$

Ex. 16. Which of the following numbers is divisible by 3?

(i) 541326

(ii) 5967013

Sol. (i) Sum of digits in 541326 = $(5 + 4 + 1 + 3 + 2 + 6) = 21$, which is divisible by 3.

Hence, 541326 is divisible by 3.

(ii) Sum of digits in 5967013 = $(5 + 9 + 6 + 7 + 0 + 1 + 3) = 31$, which is not divisible by 3.

Hence, 5967013 is not divisible by 3.

Ex. 17. What least value must be assigned to * so that the number 197*5462 is divisible by 9?

Sol. Let the missing digit be $x.$

Sum of digits = $(1 + 9 + 7 + x + 5 + 4 + 6 + 2) = (34 + x).$

For $(34 + x)$ to be divisible by 9, x must be replaced by 2.

Hence, the digit in place of * must be 2.

Ex. 18. Which of the following numbers is divisible by 4?

(i) 67920594

(ii) 618703572

Sol. (i) The number formed by the last two digits in the given number is 94, which is not divisible by 4.

Hence, 67920594 is not divisible by 4.

(ii) The number formed by the last two digits in the given number is 72, which is divisible by 4.

Hence, 618703572 is divisible by 4.